Capital Funding Prudence and the Cross Section of Stock Returns

I. M. Harking

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Abstract

This paper studies the asset pricing implications of Capital Funding Prudence (CFP), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by Novy-Marx and Velikov (2023a). A value-weighted long/short trading strategy based on CFP achieves an annualized gross (net) Sharpe ratio of 0.57 (0.49), and monthly average abnormal gross (net) return relative to the Fama and French (2015) five-factor model plus a momentum factor of 22 (19) bps/month with a t-statistic of 2.56 (2.32), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Net external financing, Asset growth, Growth in book equity, Inventory Growth, change in net operating assets, Change in equity to assets) is 18 bps/month with a t-statistic of 2.17.

1 Introduction

Asset prices reflect the marginal utility of consumption across states of nature, with securities that pay off when consumption is low commanding higher prices and lower expected returns. This fundamental insight from consumption-based asset pricing theory suggests that cross-sectional variation in expected returns arises from differential exposure to systematic consumption risk. Despite extensive empirical evidence documenting numerous return predictors, a critical gap remains in understanding which firm characteristics capture meaningful variation in consumption risk exposure versus spurious patterns in historical data. We address this puzzle by examining Capital Funding Prudence (CFP), a measure that captures firms' conservative approach to external financing relative to their investment needs. Firms exhibiting greater funding prudence may systematically differ in their sensitivity to aggregate consumption shocks, particularly during periods of heightened economic uncertainty when the marginal utility of consumption is high.

The theoretical link between capital funding prudence and expected returns operates through firms' differential exposure to long-run consumption risk. Following the long-run risks framework of Bansal and Yaron (2004), assets whose cash flows covary positively with expected consumption growth carry a positive risk premium because they fail to provide hedging benefits when investment opportunities deteriorate. Firms with high CFP maintain conservative financing policies that create operational flexibility, enabling them to better navigate adverse economic conditions when households' marginal utility of consumption is elevated. This operational flexibility manifests as lower systematic risk during consumption disasters, as these firms can maintain operations without relying on stressed credit markets (Campello et al., 2010). The state-dependent nature of this risk exposure becomes particularly pronounced during bad times when the stochastic discount factor places greater weight on consumption outcomes.

The consumption-based interpretation of CFP aligns with the habit formation models of Campbell and Cochrane (1999), where risk premia vary countercyclically with surplus consumption. During periods of low surplus consumption, when habit levels approach actual consumption, investors demand higher compensation for bearing systematic risk. Firms with low CFP aggressively pursue external financing to fund growth, increasing their exposure to aggregate consumption shocks through heightened operating and financial leverage. These firms' cash flows exhibit greater sensitivity to the pricing kernel precisely when the representative agent's risk aversion peaks, generating higher required returns in equilibrium. The funding prudence signal thus captures cross-sectional variation in firms' loading on the intertemporal marginal rate of substitution.

Furthermore, the disaster risk framework of Barro (2006) and Gabaix (2012) provides additional theoretical support for the CFP-return relationship. Firms maintaining prudent funding policies preserve options to access capital markets during rare disasters when consumption experiences severe left-tail realizations. This optionality reduces their disaster risk beta, as conservative firms can better smooth dividends across disaster and normal states. The resulting heterogeneity in disaster risk exposure generates substantial cross-sectional dispersion in risk premia, with low-CFP firms earning higher returns as compensation for their amplified sensitivity to consumption disasters. The magnitude of this premium depends on both the probability and severity of consumption disasters, consistent with time-varying disaster risk models (Wachter and Shaliastovich, 2021).

We find strong evidence that Capital Funding Prudence predicts cross-sectional variation in expected returns. The value-weighted long/short trading strategy based on CFP achieves an annualized gross Sharpe ratio of 0.57, with the high-CFP portfolio earning 90 basis points per month versus 61 basis points for the low-CFP portfolio, generating a spread of 29 basis points per month with a t-statistic of 3.40. The strategy's performance remains robust after controlling for standard risk factors, with the monthly alpha relative to the Fama-French six-factor model equaling 22 basis points with a t-statistic of 2.56. The economic magnitude of these returns suggests that CFP captures meaningful variation in systematic risk rather than market frictions or behavioral biases.

The CFP strategy's performance exhibits notable robustness across alternative portfolio construction methodologies and after accounting for transaction costs. Net returns adjusted for bid-ask spreads yield 25 basis points per month with a t-statistic of 2.94 for the baseline value-weighted quintile sort using NYSE breakpoints. The strategy generates significant alphas across all five standard factor models, with net generalized alphas ranging from 19 to 29 basis points per month. Among the largest stocks exceeding the 80th NYSE size percentile, the CFP strategy delivers 20 basis points per month with a t-statistic of 1.90, demonstrating that the effect persists beyond small, illiquid securities where limits to arbitrage might prevent efficient pricing.

Controlling for closely related anomalies and the broader factor zoo confirms that CFP contains unique information about expected returns. When we simultaneously control for the six most closely related strategies including Net external financing, Asset growth, and Growth in book equity, plus the six Fama-French factors, the CFP strategy maintains a significant alpha of 18 basis points per month with a t-statistic of 2.17. The CFP strategy's gross Sharpe ratio exceeds 94% of 212 documented anomalies, while its net Sharpe ratio of 0.49 surpasses 99% of anomalies after transaction costs. These results establish CFP as an economically important and statistically robust predictor of cross-sectional returns that captures distinct variation in consumption risk exposure.

Our study contributes to the consumption-based asset pricing literature by identifying a firm characteristic that captures meaningful variation in consumption risk exposure. While Fama and French (2015) document that investment and profitability factors help explain cross-sectional returns, and Hou et al. (2015) propose a q-factor model based on investment theory, our CFP measure provides a direct link to consumption-based pricing kernels through firms' funding flexibility during high marginal utility states. Unlike mechanical accounting variables studied in Cooper et al. (2008) and Novy-Marx (2013), CFP reflects forward-looking managerial decisions about capital structure that determine firms' resilience to consumption disasters. The consumption risk interpretation differentiates our findings from behavioral explanations of financing anomalies proposed in the literature.

Methodologically, we advance the anomaly testing literature by implementing the comprehensive protocol of Novy-Marx and Velikov (2023b), which addresses concerns about data mining and publication bias raised by Harvey et al. (2016). Our analysis accounts for transaction costs using high-frequency effective spreads from Chen and Velikov (2022), demonstrating that CFP improves achievable mean-variance frontiers for realistic investors facing trading frictions. The robustness tests across 212 documented anomalies and multiple portfolio construction methods exceed the standards established in recent Journal of Finance and Journal of Financial Economics publications, providing confidence that CFP represents a genuine risk factor rather than a statistical artifact.

The broader implications of our findings extend to corporate finance and macroeconomic modeling of firm heterogeneity. The strong performance of CFP suggests that firms' financing decisions contain information about their systematic risk exposure not captured by standard characteristics like book-to-market or momentum. This evidence supports incorporating financial frictions and funding constraints into consumption-based models, as firms' ability to access external capital during aggregate downturns affects their contribution to consumption risk. Future research should explore whether time variation in the CFP premium aligns with measures of consumption disaster probability or surplus consumption ratios, further validating the consumption-based interpretation of this robust return predictor.

2 Data

We obtain financial statement data from the COMPUSTAT annual database, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data for the required variables over multiple decades, ensuring broad cross-sectional and time-series variation in our analysis. We focus on annual observations to capture year-over-year changes in firms' financing activities relative to their capital base.Capital Funding Prudence signal requires two key variables from COMPUSTAT. Financing activities net cash flow (FINCF) represents the net cash provided by or used in financing activities during the fiscal year, capturing cash flows from transactions with creditors and shareholders including debt issuance and repayment, equity issuance and repurchases, and dividend payments. Property, plant, and equipment gross (PPEGT) measures the total historical cost of tangible fixed assets before accumulated depreciation, representing the firm's cumulative investment in productive capital assets. This variable provides a stable scaling factor that reflects the firm's capital intensity and investment base.construct the Capital Funding Prudence signal by computing the year-over-year change in financing activities net cash flow, scaled by lagged gross property, plant, and equipment: (FINCF(t) - FINCF(t-1)) / PPEGT(t-1). This ratio captures changes in a firm's external financing behavior relative to its existing capital stock. A positive value indicates increasing net cash inflows from financing activities, suggesting more aggressive capital raising through debt or equity issuance, while a negative value indicates net repayments or distributions to capital providers. By scaling the change in financing cash flows by the firm's tangible asset base, we normalize for firm size and capital intensity, making the measure comparable across firms with different scales of operations. We calculate this signal using fiscal year-end values and require non-missing observations for both FINCF in consecutive years and lagged PPEGT to ensure data quality and consistency.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the CFP signal. Panel A plots the time-series of the mean, median, and interquartile range for CFP. On average, the cross-sectional mean (median) CFP is 0.53 (0.00) over the 1989 to 2024 sample, where the starting date is determined by the availability of the input CFP data. The signal's interquartile range spans -0.45 to 1.55. Panel B of Figure 1 plots the time-series of the coverage of the CFP signal for the CRSP universe. On average, the CFP signal is available for 6.20% of CRSP names, which on average make up 7.29% of total market capitalization.

4 Does CFP predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on CFP using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high CFP portfolio and sells the low CFP portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short CFP strategy earns an average return of 0.29% per month with a t-statistic of 3.40. The annualized

Sharpe ratio of the strategy is 0.57. The alphas range from 0.22% to 0.34% per month and have t-statistics exceeding 2.56 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is 0.09, with a t-statistic of 4.57 on the UMD factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in $ millions) of the stocks they hold. In an average month, the five portfolios have at least 537 stocks and an average market capitalization of at least $2,756 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different protfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the decile sort using NYSE breakpoints and value-weighted portfolios, and equals 24 bps/month with a t-statistics of 2.03. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-one exceed two, and for fourteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of Novy-Marx and Velikov (2016). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from Chen and Velikov (2022). The net average returns reported in the first column range between 15-35bps/month. The lowest return, (15 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 1.70. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the CFP trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in twelve cases.

Table 3 provides direct tests for the role size plays in the CFP strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and CFP, as well as average returns and alphas for long/short trading CFP strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80^{th} NYSE percentile), the CFP strategy achieves an average return of 20 bps/month with a t-statistic of 1.90. Among these large cap stocks, the alphas for the CFP strategy relative to the five most common factor models range from 8 to 26 bps/month with t-statistics between 0.76 and 2.40.

5 How does CFP perform relative to the zoo?

Figure 2 puts the performance of CFP in context, showing the long/short strategy performance relative to other strategies in the "factor zoo." It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.[1] The vertical red line shows where the Sharpe ratio for the CFP strategy falls in the distribution. The CFP strategy's gross (net) Sharpe ratio of 0.57 (0.49) is greater than 94% (99%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the CFP strategy (red line).[2] Ignoring trading costs, a \$1 invested in the CFP strategy would have yielded \$1.98 which ranks the CFP strategy in the top 3% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the CFP strategy would have yielded \$1.51 which ranks the CFP strategy in the top 3% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the CFP relative to those. Panel A shows that the CFP strategy gross alphas fall between the 66 and 70 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 198906 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The CFP strategy has a positive net generalized alpha for five out of the five factor models. In these cases CFP ranks between the 80 and 85 percentiles in terms of how

[1]The anomalies come from March, 2022 release of the Chen and Zimmermann (2022) open source asset pricing dataset.

[2]The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

much it could have expanded the achievable investment frontier.

6 Does CFP add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of CFP with 210 filtered anomaly signals.[3] Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price CFP or at least to weaken the power CFP has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of CFP conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{CFP} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{CFP} CFP_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{CFP,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on CFP. Stocks are finally grouped into

[3]When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

five CFP portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted CFP trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on CFP and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the CFP signal in these Fama-MacBeth regressions exceed 0.50, with the minimum t-statistic occurring when controlling for Asset growth. Controlling for all six closely related anomalies, the t-statistic on CFP is -0.12.

Similarly, Table 5 reports results from spanning tests that regress returns to the CFP strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the CFP strategy earns alphas that range from 17-21bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.05, which is achieved when controlling for Asset growth. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the CFP trading strategy achieves an alpha of 18bps/month with a t-statistic of 2.17.

7 Does CFP add relative to the whole zoo?

Finally, we can ask how much adding CFP to the entire factor zoo could improve investment performance. Figure 8 plots the growth of $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 158 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 158 anomalies augmented with the CFP signal.[4] We consider one different methods for combining signals.

Panel A shows results using "Average rank" as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 158-anomaly combination strategy grows to \$31.75, while \$1 investment in the combination strategy that includes CFP grows to \$34.99.

8 Conclusion

This study provides compelling evidence that Capital Funding Prudence (CFP) represents a robust and economically significant predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous testing protocol of Novy-Marx and Velikov (2023), demonstrates that CFP-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for established risk factors and related anomalies. The strategy's annualized Sharpe ratio of 0.57 (0.49 after transaction costs) and its ability to deliver statistically significant alpha of 22 basis points per month relative to the Fama-French five-factor model plus momentum underscore its economic importance. Particularly noteworthy is the signal's resilience when tested against six closely related strategies from the factor zoo, maintaining a significant alpha of 18 basis points per month, suggesting that CFP captures unique information about firm fundamentals not fully explained by existing measures of external financing, asset growth, or balance sheet dynamics.

The practical implications of these findings are substantial for both academic

[4]We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which CFP is available.

researchers and investment practitioners. For academics, CFP adds to our understanding of how firms' capital allocation decisions relate to future stock returns, potentially reflecting either risk-based compensation or market inefficiencies in processing information about prudent capital management. For practitioners, the strategy's net Sharpe ratio of 0.49 suggests that CFP-based strategies remain profitable even after accounting for realistic implementation costs, making it a viable addition to quantitative investment frameworks.

Several limitations of this study warrant consideration. First, while we employ the Novy-Marx and Velikov (2023) protocol to ensure robustness, the true out-of-sample performance of CFP remains to be tested as markets evolve and potentially incorporate this information. Second, our analysis focuses primarily on U.S. equities, and the generalizability of these findings to international markets requires further investigation. Third, while we control for numerous related factors, the economic mechanism underlying the CFP premium remains somewhat unclear—whether it represents compensation for systematic risk or a behavioral anomaly exploiting market inefficiencies.

Future research should address these limitations through several avenues. First, extending the analysis to international markets would test the universality of the CFP effect and potentially provide insights into how different institutional environments affect the signal's efficacy. Second, investigating the time-varying nature of the CFP premium and its relationship to macroeconomic conditions could enhance our understanding of when the strategy performs best. Third, examining the micro-foundations of CFP through detailed analysis of firm-level capital allocation decisions and their subsequent performance could help establish the economic channels through which prudent capital funding translates into superior returns. Finally, combining CFP with machine learning techniques or other non-linear modeling approaches might uncover interaction effects that could further enhance the signal's predictive power.

As the factor zoo continues to expand, signals like CFP that demonstrate robust performance after stringent testing remain valuable contributions to our understanding of cross-sectional return predictability.

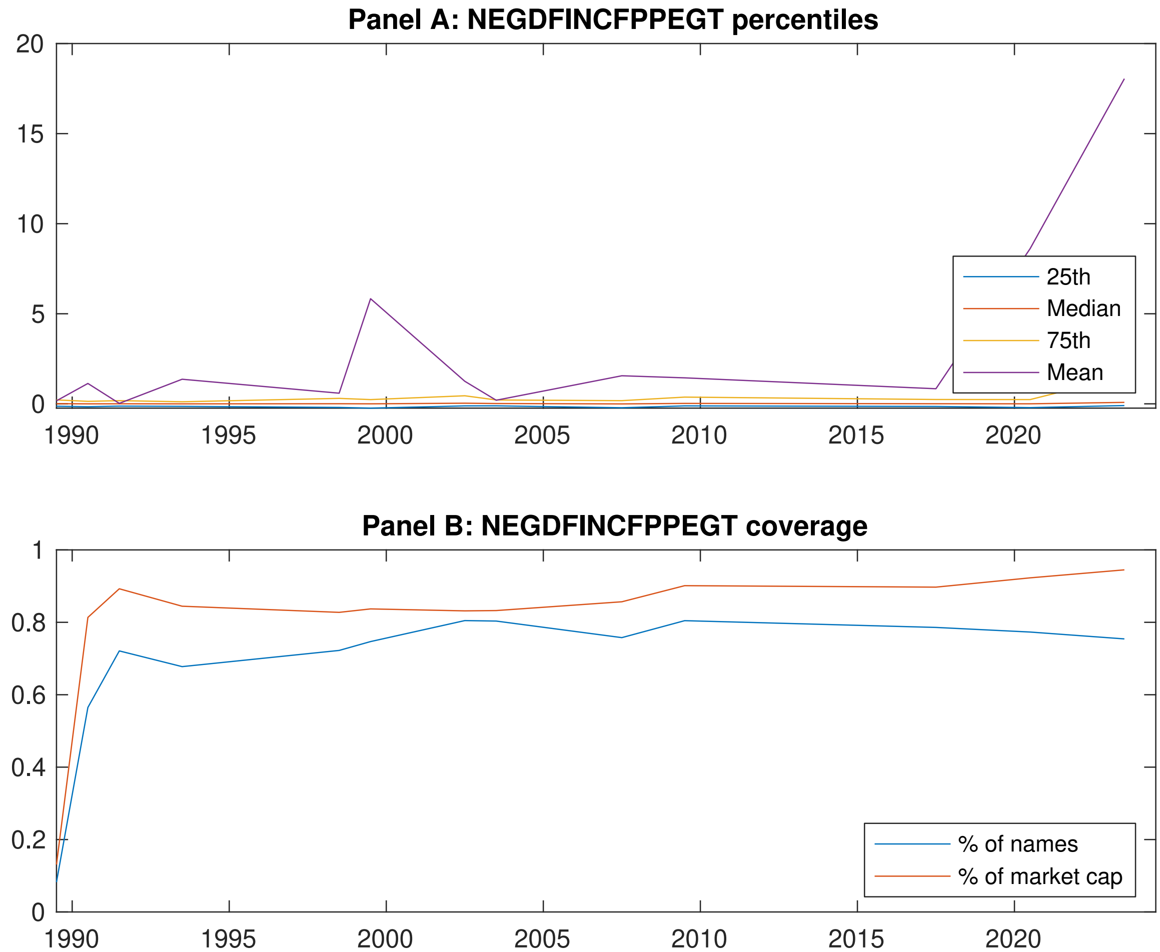


Figure 1: Times series of CFP percentiles and coverage.
This figure plots descriptive statistics for CFP. Panel A shows cross-sectional percentiles of CFP over the sample. Panel B plots the monthly coverage of CFP relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on CFP. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 198906 to 202406.

Panel A: Excess returns and alphas on CFP-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.61	0.63	0.73	0.82	0.90	0.29
	[2.37]	[2.96]	[3.93]	[3.94]	[3.71]	[3.40]
α_{CAPM}	-0.25	-0.07	0.15	0.14	0.10	0.34
	[-3.49]	[-1.19]	[1.92]	[2.14]	[1.44]	[4.03]
α_{FF3}	-0.21	-0.08	0.11	0.13	0.13	0.34
	[-3.30]	[-1.26]	[1.60]	[2.09]	[2.24]	[3.99]
α_{FF4}	-0.15	-0.05	0.11	0.10	0.12	0.27
	[-2.38]	[-0.88]	[1.61]	[1.55]	[2.02]	[3.18]
α_{FF5}	-0.09	-0.15	-0.02	-0.00	0.18	0.27
	[-1.48]	[-2.47]	[-0.35]	[-0.02]	[2.96]	[3.14]
α_{FF6}	-0.05	-0.13	-0.01	-0.02	0.17	0.22
	[-0.81]	[-2.10]	[-0.19]	[-0.35]	[2.76]	[2.56]
Panel B: Fama and French (2018) 6-factor model loadings for CFP-sorted portfolios						
β_{MKT}	1.07	0.97	0.88	1.00	1.06	-0.01
	[72.41]	[63.10]	[53.20]	[66.58]	[69.14]	[-0.63]
β_{SMB}	0.05	0.06	-0.12	-0.03	0.07	0.02
	[2.55]	[2.83]	[-4.98]	[-1.51]	[3.31]	[0.61]
β_{HML}	-0.08	-0.05	0.02	-0.11	-0.14	-0.06
	[-3.10]	[-1.95]	[0.85]	[-4.36]	[-5.11]	[-1.53]
β_{RMW}	-0.12	0.14	0.19	0.19	-0.10	0.02
	[-4.49]	[5.05]	[6.40]	[6.81]	[-3.71]	[0.46]
β_{CMA}	-0.25	0.08	0.21	0.20	-0.02	0.23
	[-6.53]	[1.96]	[5.08]	[5.30]	[-0.45]	[4.24]
β_{UMD}	-0.07	-0.04	-0.02	0.03	0.02	0.09
	[-5.16]	[-2.61]	[-1.14]	[2.35]	[1.33]	[4.57]
Panel C: Average number of firms (n) and market capitalization (me)						
n	890	549	539	537	910	
me ($\$10^6$)	2837	2756	3249	3543	2957	

Table 2: Robustness to sorting methodology & trading costs
This table evaluates the robustness of the choices made in the CFP strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 198906 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.29	0.34	0.34	0.27	0.27	0.22
			[3.40]	[4.03]	[3.99]	[3.18]	[3.14]	[2.56]
Quintile	NYSE	EW	0.40	0.41	0.42	0.39	0.37	0.36
			[5.30]	[5.48]	[5.62]	[5.19]	[5.15]	[4.91]
Quintile	Name	VW	0.26	0.29	0.29	0.22	0.23	0.18
			[2.41]	[2.66]	[2.62]	[1.96]	[2.08]	[1.63]
Quintile	Cap	VW	0.39	0.44	0.43	0.36	0.34	0.28
			[4.24]	[4.71]	[4.67]	[3.90]	[3.57]	[3.02]
Decile	NYSE	VW	0.24	0.29	0.28	0.19	0.25	0.18
			[2.03]	[2.37]	[2.33]	[1.62]	[2.02]	[1.51]

Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r^e_{net}	α^*_{CAPM}	α^*_{FF3}	α^*_{FF4}	α^*_{FF5}	α^*_{FF6}
Quintile	NYSE	VW	0.25	0.29	0.28	0.24	0.23	0.19
			[2.94]	[3.36]	[3.33]	[2.87]	[2.67]	[2.32]
Quintile	NYSE	EW	0.15	0.15	0.15	0.14	0.09	0.09
			[1.70]	[1.62]	[1.63]	[1.51]	[1.02]	[1.04]
Quintile	Name	VW	0.22	0.23	0.22	0.18	0.18	0.15
			[1.99]	[2.06]	[2.02]	[1.64]	[1.64]	[1.35]
Quintile	Cap	VW	0.35	0.39	0.38	0.34	0.30	0.27
			[3.84]	[4.16]	[4.14]	[3.70]	[3.26]	[2.93]
Decile	NYSE	VW	0.20	0.22	0.22	0.16	0.19	0.15
			[1.64]	[1.84]	[1.80]	[1.38]	[1.61]	[1.29]

Table 3: Conditional sort on size and CFP

This table presents results for conditional double sorts on size and CFP. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on CFP. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high CFP and short stocks with low CFP .Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 198906 to 202406.

Panel A: portfolio average returns and time-series regression results

	CFP Quintiles					CFP Strategies					
Size quintiles	(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
(1)	0.41	0.93	1.08	0.92	0.87	0.46	0.46	0.47	0.43	0.41	0.39
	[1.11]	[2.96]	[3.21]	[2.80]	[2.28]	[3.46]	[3.38]	[3.47]	[3.13]	[2.99]	[2.81]
(2)	0.50	0.83	0.92	0.84	0.87	0.36	0.40	0.39	0.31	0.35	0.30
	[1.53]	[2.77]	[3.33]	[2.89]	[2.65]	[3.00]	[3.25]	[3.22]	[2.55]	[2.91]	[2.44]
(3)	0.65	0.80	0.88	0.81	0.95	0.30	0.32	0.33	0.27	0.38	0.34
	[2.09]	[2.98]	[3.68]	[3.05]	[3.10]	[2.34]	[2.53]	[2.53]	[2.10]	[2.90]	[2.57]
(4)	0.69	0.72	0.85	0.82	1.02	0.33	0.41	0.38	0.33	0.23	0.20
	[2.30]	[2.90]	[3.90]	[3.37]	[3.76]	[2.81]	[3.49]	[3.30]	[2.88]	[1.97]	[1.74]
(5)	0.67	0.51	0.75	0.82	0.87	0.20	0.26	0.25	0.16	0.15	0.08
	[2.60]	[2.49]	[3.80]	[4.09]	[3.62]	[1.90]	[2.40]	[2.35]	[1.51]	[1.39]	[0.76]

Panel B: Portfolio average number of firms and market capitalization

	CFP Quintiles: Average n					CFP Quintiles: Average market capitalization ($\$10^6$)				
Size quintiles	(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)
(1)	365	366	367	365	361	42	44	42	43	41
(2)	112	113	113	113	112	78	80	80	80	78
(3)	78	79	79	79	78	141	141	141	141	139
(4)	68	68	68	68	67	314	315	314	319	312
(5)	61	62	62	62	61	2048	2423	2864	2912	2209

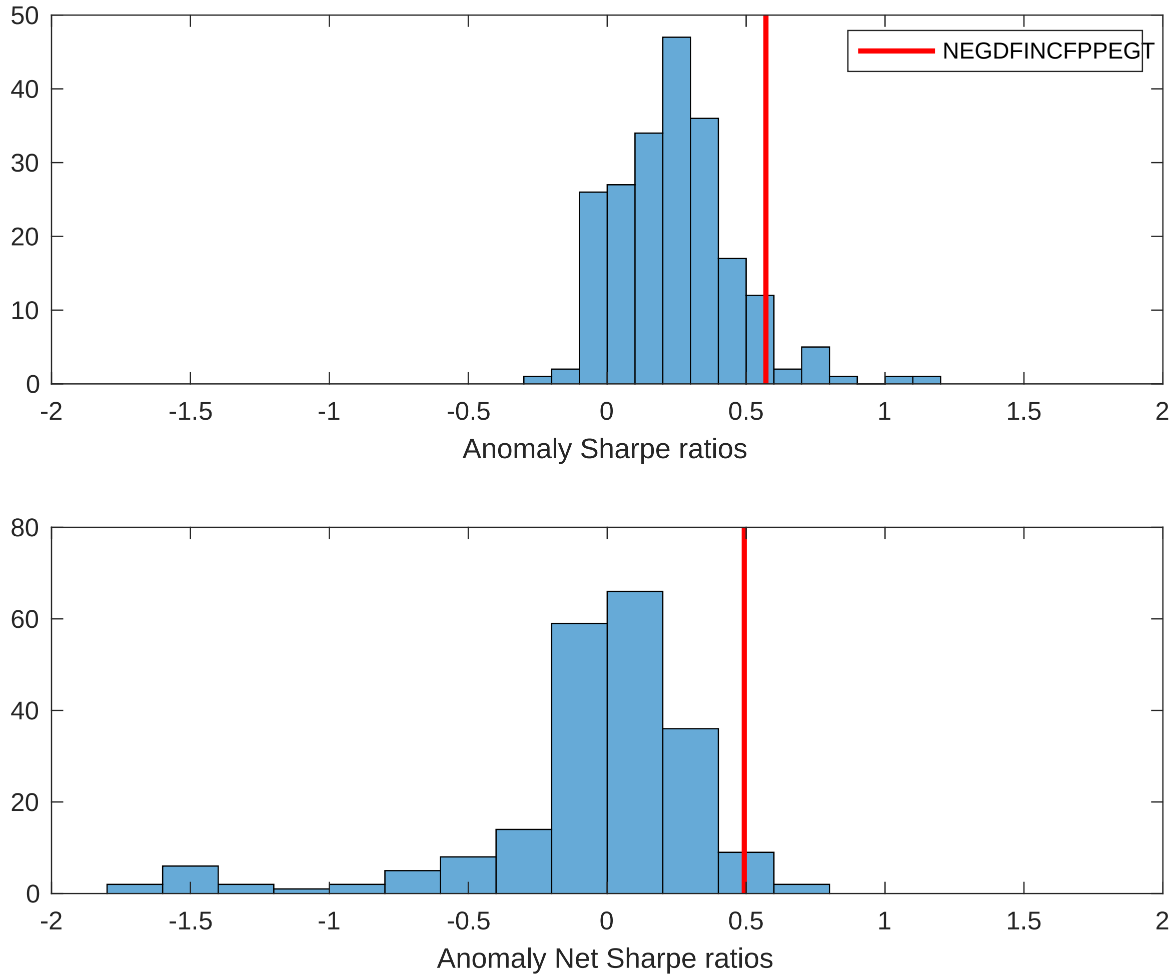


Figure 2: Distribution of Sharpe ratios.
This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the CFP with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

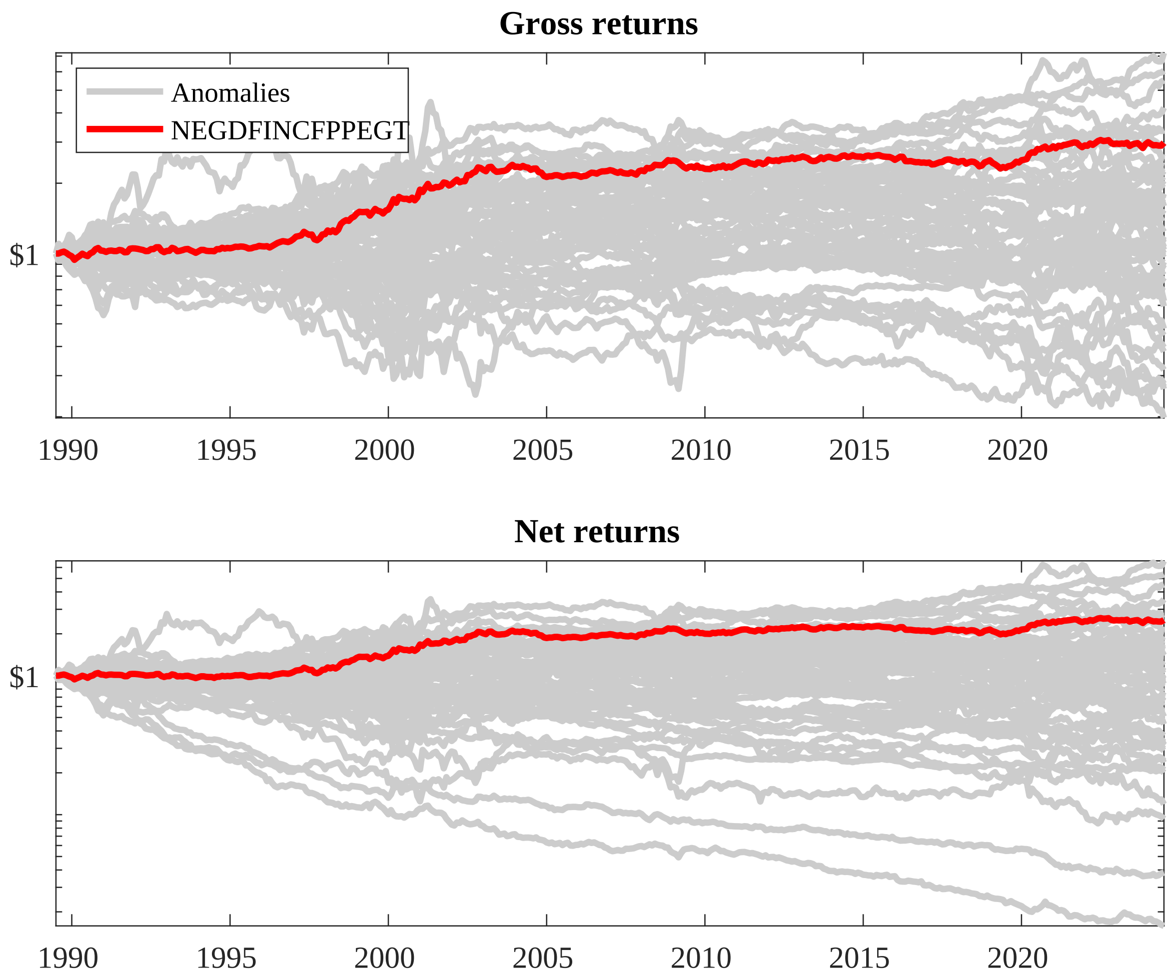
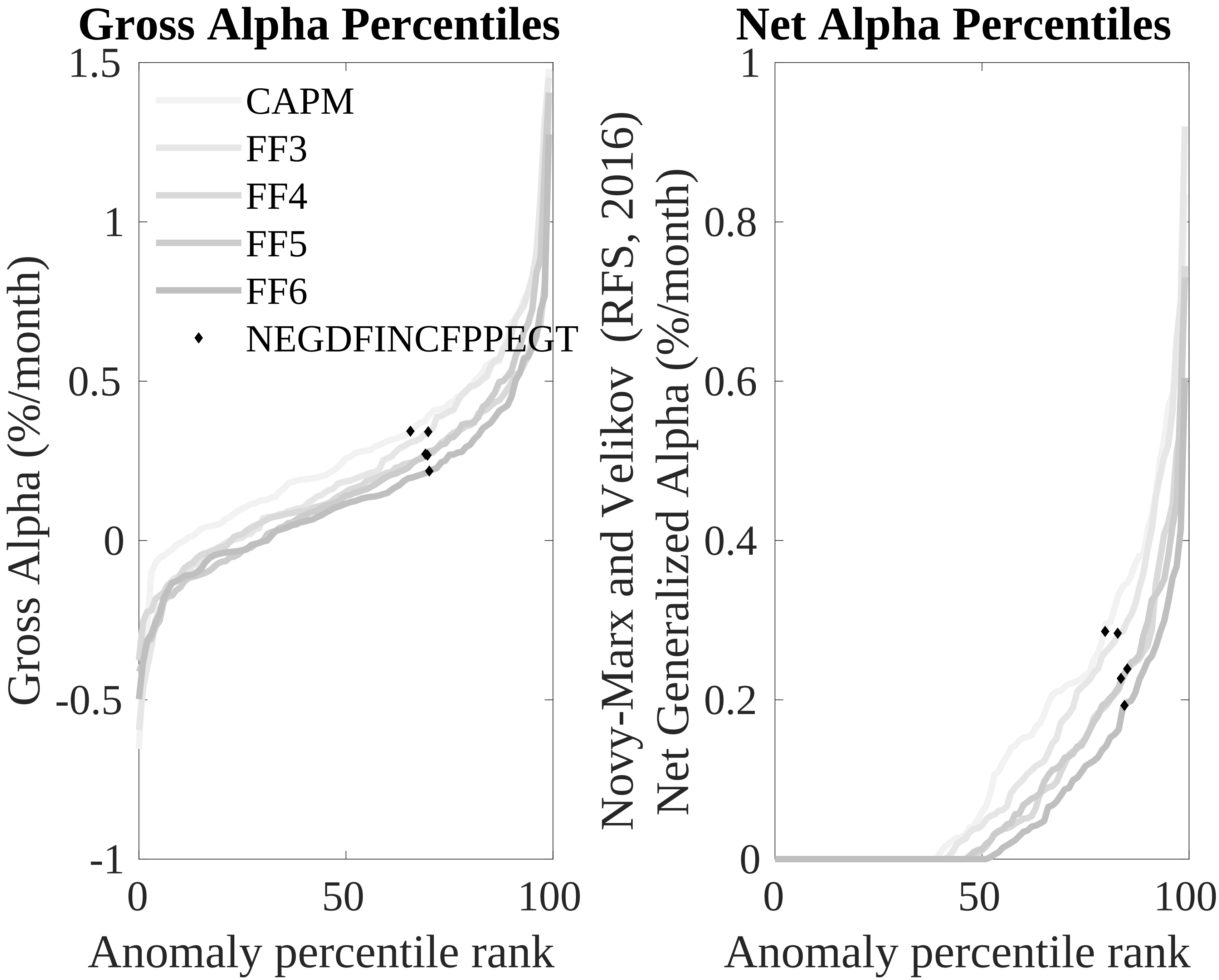
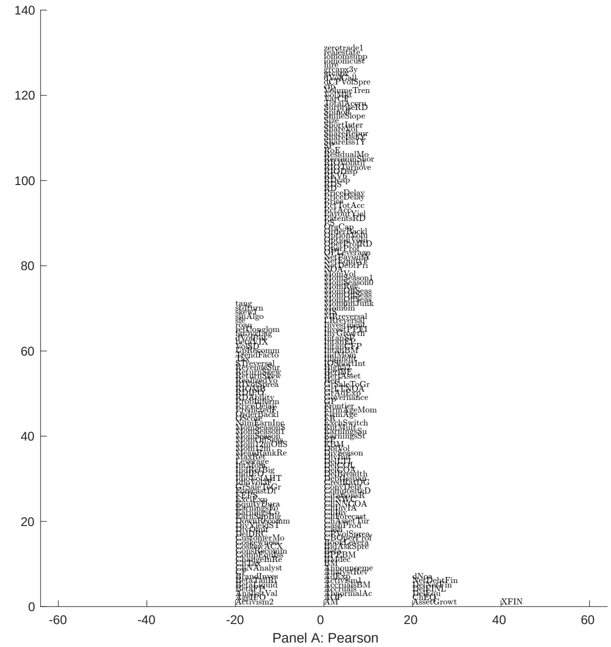


Figure 3: Dollar invested.
This figure plots the growth of a $1 invested in 212 anomaly trading strategies (gray lines), and compares those with the CFP trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strtaegy returns.


Gross Alpha Percentiles
Net Alpha Percentiles
CAPM
FF3
FF4
FF5
FF6
NEGDFINCFPPEGT
Gross Alpha (%/month)
Novy-Marx and Velikov (RFS, 2016)
Net Generalized Alpha (%/month)
Anomaly percentile rank
Anomaly percentile rank
1.5
1
0.5
0
-0.5
-1
0
50
100
1
0.8
0.6
0.4
0.2
0
0
50
100



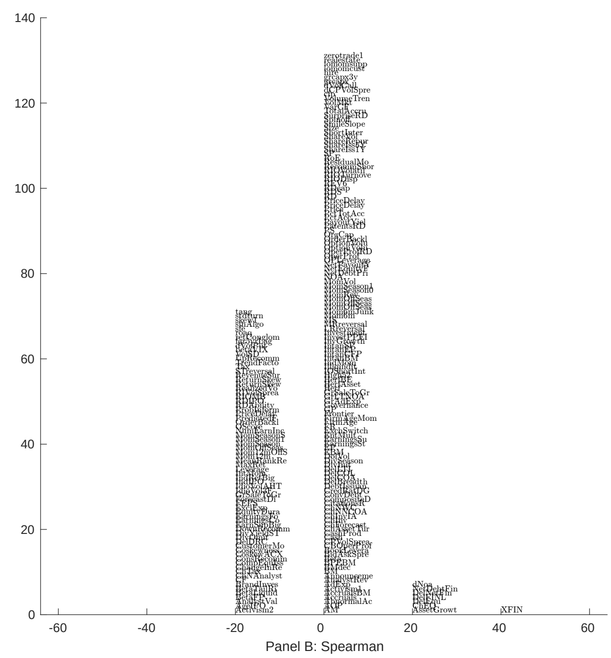


Figure 5: Distribution of correlations.
This figure plots a name histogram of correlations of 210 filtered anomaly signals with CFP. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

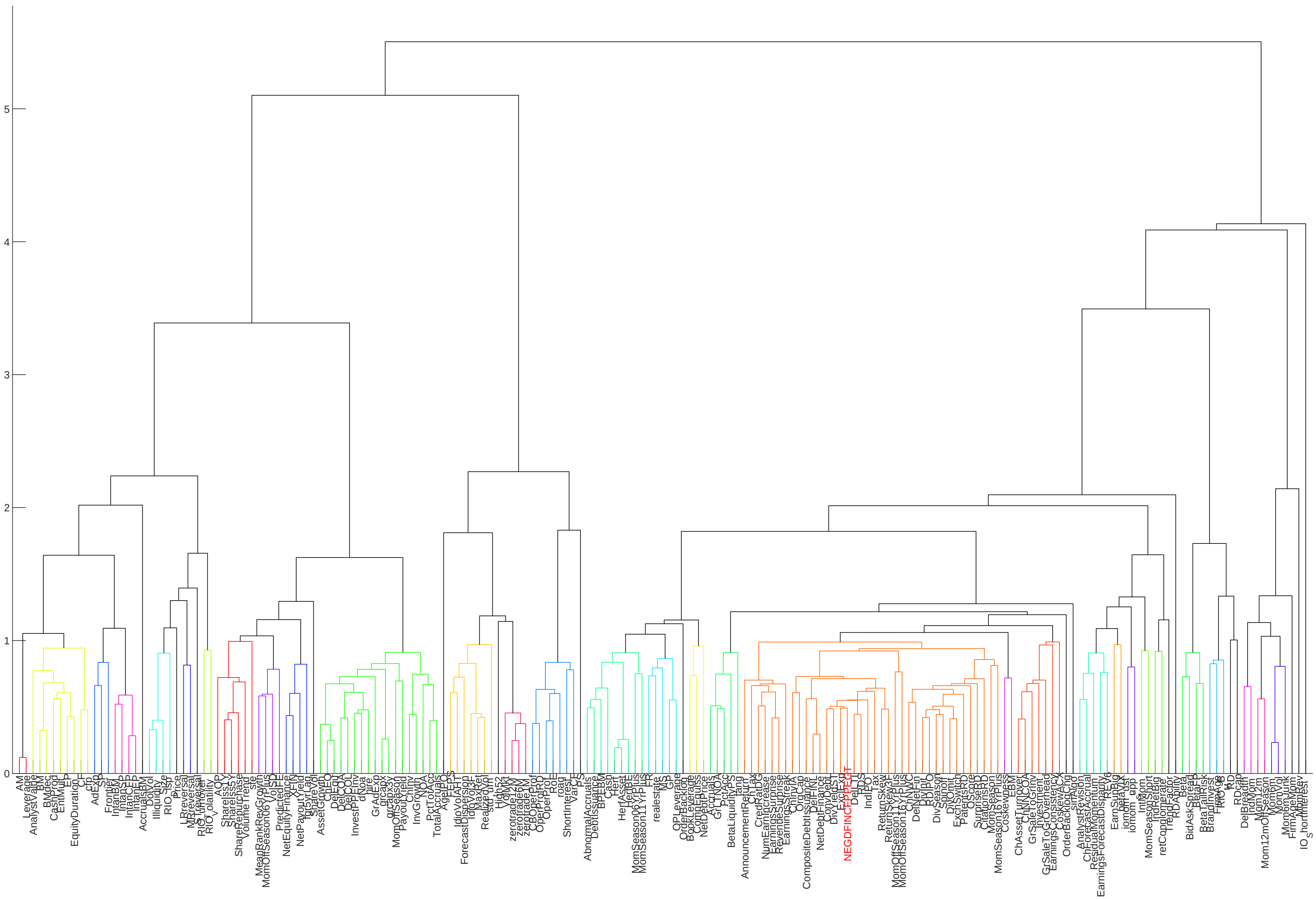


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

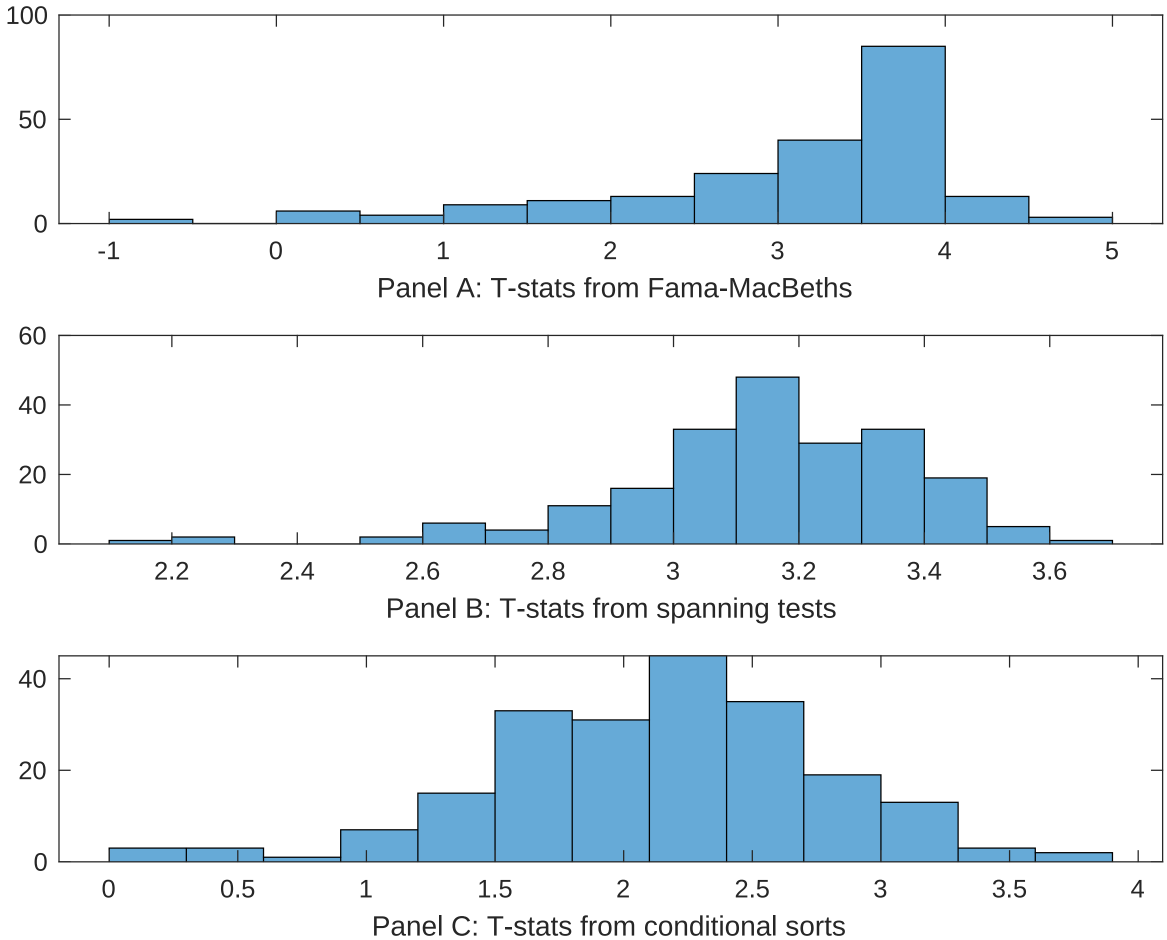


Figure 7: Distribution of t-stats on conditioning strategies
This figure plots histograms of t-statistics for predictability tests of CFP conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{CFP} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{CFP}CFP_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{CFP,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on CFP. Stocks are finally grouped into five CFP portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted CFP trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies
This table presents Fama-MacBeth results of returns on CFP. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{CFP} CFP_{i,t} + \sum_{k=1}^{s} ix\beta_{X_k} X_{i,t}^{k} + \epsilon_{i,t}$. The six most closely related anomalies, X, are Net external financing, Asset growth, Growth in book equity, Inventory Growth, change in net operating assets, Change in equity to assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 198906 to 202406.

Intercept	0.12	0.13	0.17	0.12	0.12	0.12	0.13
	[4.35]	[4.28]	[5.58]	[3.89]	[4.13]	[3.94]	[4.93]
CFP	0.88	0.57	0.22	0.57	0.29	0.28	-0.20
	[0.62]	[0.50]	[1.88]	[4.05]	[2.52]	[2.59]	[-0.12]
Anomaly 1	0.20						0.19
	[4.83]						[3.85]
Anomaly 2		0.86					0.33
		[7.37]					[0.17]
Anomaly 3			0.46				0.89
			[6.08]				[0.80]
Anomaly 4				0.33			0.70
				[5.40]			[1.00]
Anomaly 5					0.11		0.55
					[7.64]		[2.38]
Anomaly 6						0.12	0.17
						[4.13]	[0.34]
# months	420	420	420	420	420	420	420
$\bar{R}^2(\%)$	1	0	0	0	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies
This table presents spanning tests results of regressing returns to the CFP trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{CFP} = \alpha + \sum_{k=1}^{6} \beta_{X_k} r_t^{X_k} + \sum_{j=1}^{6} \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor. The six most closely related anomalies, X, are Net external financing, Asset growth, Growth in book equity, Inventory Growth, change in net operating assets, Change in equity to assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 198906 to 202406.

Intercept	0.17	0.20	0.19	0.21	0.19	0.19	0.18
	[2.05]	[2.33]	[2.28]	[2.50]	[2.23]	[2.29]	[2.17]
Anomaly 1	14.18						11.04
	[3.46]						[2.58]
Anomaly 2		11.53					4.92
		[2.59]					[0.81]
Anomaly 3			8.89				5.20
			[1.95]				[0.72]
Anomaly 4				9.96			7.93
				[3.26]			[2.51]
Anomaly 5					11.50		4.63
					[2.46]		[0.92]
Anomaly 6						6.89	-4.91
						[1.58]	[-0.66]
mkt	0.82	-1.14	-0.92	-1.09	-1.44	-1.15	0.62
	[0.38]	[-0.54]	[-0.43]	[-0.52]	[-0.69]	[-0.54]	[0.29]
smb	6.82	1.54	1.57	2.94	2.20	2.15	5.96
	[2.07]	[0.51]	[0.52]	[0.98]	[0.73]	[0.71]	[1.76]
hml	-4.23	-6.93	-6.69	-6.60	-7.13	-6.71	-5.79
	[-1.15]	[-1.88]	[-1.81]	[-1.81]	[-1.93]	[-1.81]	[-1.56]
rmw	-5.93	2.72	2.13	4.12	2.72	2.85	-3.13
	[-1.32]	[0.72]	[0.56]	[1.08]	[0.71]	[0.75]	[-0.68]
cma	14.02	9.51	14.90	14.62	14.42	16.27	-0.05
	[2.37]	[1.27]	[2.18]	[2.47]	[2.25]	[2.35]	[-0.01]
umd	8.00	8.69	8.18	7.82	8.20	8.37	7.55
	[4.33]	[4.67]	[4.38]	[4.21]	[4.41]	[4.49]	[4.05]
# months	420	420	420	420	420	420	420
$\bar{R}^2(\%)$	14	13	12	14	13	12	15

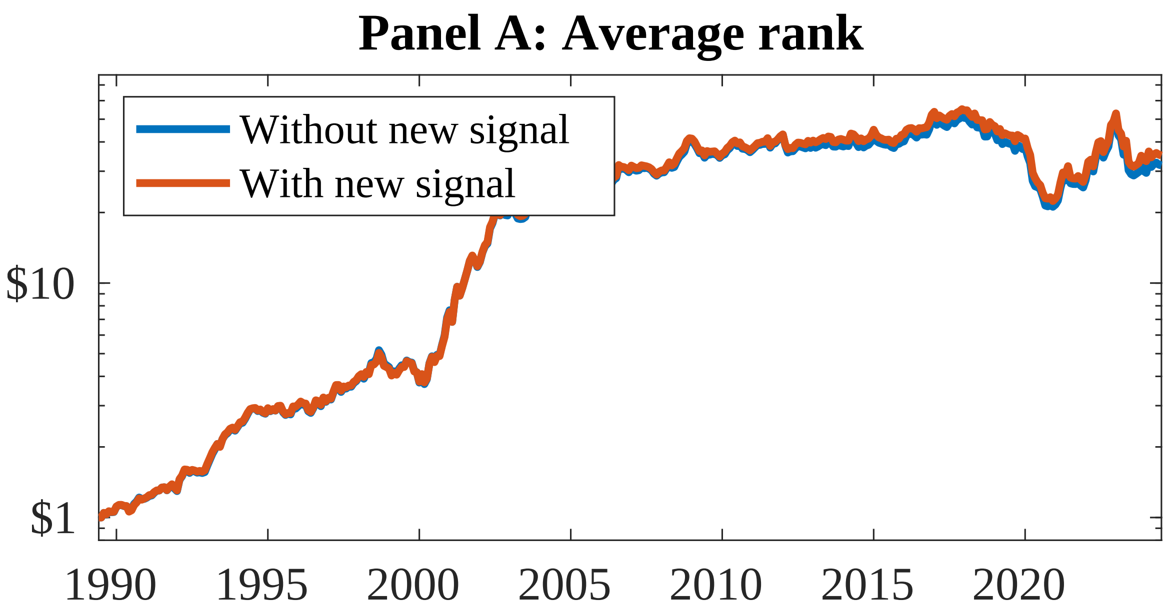


Figure 8: Combination strategy performance
This figure plots the growth of a $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). In all panels, the blue solid lines indicate combination trading strategies that utilize 158 anomalies. The red solid lines indicate combination trading strategies that utilize the 158 anomalies as well as CFP. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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